Mert Tuna

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"Senior student double majoring in Electrical & Electronics Engineering and Computer Engineering, a dedicated lifelong learner passionate about acquiring skills and sharing knowledge. Driven by a commitment to perform best of any work, I am eager to leverage my multidisciplinary background to make meaningful contributions in professional, real-world applications."

Work Experience

Koç University Association of Space & Rocketry (KUASAR) **İstanbul, Sarıyer**
Team Member *Aug 2024 – Present*

- Contributed to the creation of two custom PCB for the rocket's flight computer, tailored for competitions including Teknofest (Turkiye) and IREC (USA).
- Collaborated the ground station UI for real-time rocket telemetry and mission data visualization, enabling seamless communication between launch control and onboard systems.

Roketsan Hardware Design Department **Ankara, Lalahan**
Intern *June 2024 – July 2024*

- Researched various laser distance technologies.
- Tested and reverse engineered OPT3101 Evolution module.
- Using Altium Designer software, designed a custom printed circuit board using OPT3101 sensor under some system considerations and constraints.

Koç University Renewable Energy Community (KUREC) **İstanbul, Sarıyer**
Teknofest Efficiency Challenge Team Member *Dec 2023 – July 2024*

- Involved in the Teknofest Efficiency Challenge, collaborating with a multidisciplinary team to design and develop key components for the Hydromobile, poised to be the cornerstone project for the community's inception.
- Tasked with the design of the motor driver, telemetry, and Vehicle Control Unit (VCU), leveraging technical expertise to contribute to the project's innovative objectives.
- Engaged in comprehensive learning of Altium Designer for the PCB design of the motor driver.

Koç University Optical Microsystems Laboratory (OML) **İstanbul, Sarıyer**
Undergraduate Researcher *Oct 2023 – Jan 2024*

- Spearheaded the adoption of Lidar Toolbox in MATLAB for the preprocessing, visualization, and analysis of lidar data, under the supervisorship of Prof. Dr. Hakan Urey and PhD student Mehrdad Khodapanhandeh.
- Specialized in the development and testing of advanced 2D and 3D SLAM algorithms, leading to a more nuanced understanding of spatial mapping technologies.
- Demonstrated expertise in object detection within point clouds, leveraging various pre-existing datasets to enhance detection accuracy and efficiency.

Education

Koç University **İstanbul/Türkiye**
Bachelor of Science *Sep 2021 – Present*
Double Major in Electrical & Electronics Engineering and Computer Engineering
Minor in Business Administration
Cumulative GPA: 3.85/4.00

Hong Kong University of Science and Technology **Sai Kung/Hong Kong**
Exchange Semester (FALL) *Aug 2025 – Dec 2025*
Electronics and Computer Engineering Department
- Deep Learning in Computer Vision
- Fundamentals of Artificial Intelligence
- Robotics, Planning, Control and Modelling
- Operating Systems

Bahçeşehir High School for Science and Technology **Samsun/Türkiye**
Valedictorian; GPA: 99.73/100 *Sep 2017 -June 2021*

Relevant Coursework

Computer Engineering - Introduction to Programming (Python) • Advanced Programming (Java, OOP) • Data Structures and Algorithms • Computer Systems and Programming (C, Assembly) • Software Engineering • Discrete Mathematics for Computer Science and Engineering • Operating Systems • Deep Learning • Fundamentals of AI
Electrical & Electronics Engineering - Signals & Systems • Systems, Control & Communication • Microelectronic Circuits and Devices • Circuits • Digital System Design • Interactive Device Design & Prototyping I & II • Introduction to Electrical and Electronics Engineering (MATLAB) • Robotics
Business Administration - Service Operations Management • Operations Strategy • Big Data in Business and Public Sector, Financial Accounting • Introduction to Financial Management • Marketing Research • Marketing Management

Projects

- **AI-Based Mental Health Text Classification:** Co-authored a comparative research study benchmarking a hybrid NLP pipeline using KNIME and Python on 51,000 text samples, achieving 83.6% accuracy by fine-tuning a BERT transformer to detect specific mental health conditions.

- **Pedestrian Action Prediction for Autonomous Driving:** Co-authored a comparative study predicting pedestrian crossing intent by designing and benchmarking three distinct modular computer vision pipelines, incorporating diverse object detection, tracking, and intention prediction models for autonomous agents.

- **Unity Junior Programmer Pathway (Certification & Portfolio):** Earned the certification, completing a intensive curriculum focused on C# scripting and professional software development workflows. Developed a portfolio **5+ technical prototypes** demonstrating mastery in Unity's physics engine, wave-spawner logic, and event-driven UI systems. Gained proficiency in debugging, performance profiling and implementation of technical documentation.

- **3D Isometric City Builder (Capstone Project):** Engineered a custom 3D isometric camera controller featuring smooth panning, rotation, and dynamic zoom functionality tailored for strategy gameplay. Applied OOP Principles—**Inheritance, Abstraction, Polymorphism, and Encapsulation**—to create clean, maintainable codebases. Engineered an autonomous vehicle navigation system for gameplay enhancement. Designed and implemented a dynamic UI tooltip system that interprets game-object data to provide real-time feedback to the player.

- **Java Tower Defense Game:** Led the development of the **top-ranked** tower defense game in a collaborative Agile environment, earning maximum bonus credits for implementing advanced and extra game mechanics and superior software architecture using Java.

- **UNO Card Game Simulation:** Engineered a fully functional digital adaptation of UNO card game as a solo project, utilizing advanced Object-Oriented Programming (OOP) principles.

- **5-Axis 3D Printer Prototype:** Designed and prototyped a functional 5-axis printer over a two-semester capstone, successfully integrating hardware mechanics with embedded software to execute complex motion paths.

Extracurricular Activities

QNB Finansbank – UPCLUB **İstanbul, Sarıyer**
Active Member *Jan 2025 – Jun 2025*

- Selected as one of 40 university students from a nationwide pool for QNB Finansbank's exclusive professional development program.
- Engaged in a practical case study on a real-world business challenge, developing analytical and problem-solving skills.
- Participated in exclusive workshops and training sessions.

Koç University - Experimental Physics Summer Camp **İstanbul, Sarıyer**
Summer Academy Mentor *July 2025 – Aug 2025*

- Mentored high school students in a laboratory setting, assisting with experimental setups and explaining fundamental physics concepts.

Koç University **İstanbul, Sarıyer**
Student Mentor *Aug 2024 – Jun 2025*

- Mentored newcomers, providing guidance on academic resources, course selection, and campus life to facilitate their transition into university, also advised on social integration to improve student retention and success.

Southern Denmark University Summer School **Odense/Denmark**
Drones for Computer Vision Applications (Passed) *Aug 2024 – Aug 2024*

- Working in a group of three with peers from diverse backgrounds, successfully completed an intensive course in two weeks: Multirotor drone architecture, Raspberry Pi camera payload, autonomous mission, drone ethics, computer vision applications on detecting flower pots on the ground (OpenCV).

Koç University KOLT Office **İstanbul, Sarıyer**
Tutor *Feb 2024 – June 2024*

- Employed by KOLT Office as a Tutor for ELEC205 (Digital Design), a course completed with distinction among the top students.
- Provided personalized assistance to students on assignments, conducted review sessions for midterm and final exam preparation.

Koç University – KUDANS Student Dance Club **İstanbul, Sarıyer**
Performance Team Member - Swing Dance *Jan 2024 – June 2024*

- Collaborated with a team of 20+ dancers to learn and perform choreographed Swing dance routines to perform on various university and public events.
- Committed approximately 7 hours per week to rehearsals and training, demonstrating strong time management alongside academic responsibilities.

Koç University **İstanbul, Sarıyer**
Section Leader *Oct 2023 – Jan 2024*

- Selected for a section leader role in COMP125 (Introduction to Programming) course, following an outstanding performance in the previous semester, provided comprehensive support to students during lab hours and individualized office hours, aiding in the clarification of complex concepts and enhancing their hands-on experience.

Honors & Awards

Vehbi Koç Honor Award | Koç University (5 Semesters)

Awarded for achieving a GPA above 3.75 in: Fall 2022, Spring 2023, Spring 2024, Fall 2024, Spring 2025

Dean's Honor List | Koç University (Fall 2023)

Skills & Interests

Technical Skills

- **Programming Languages:** Python, Java, C, MATLAB, Assembly**.**
- **AI & Computer Vision:** PyTorch, NumPy, Deep Learning, Object Detection, Agentic AI, KNIME Analytics Platform.
- **Software Engineering:** Object-Oriented Programming (OOP), Git, Unit Testing, Design Patterns, Agile/Scrum Methodologies.
- **Hardware, Electronics & Embedded Systems:** Altium Designer (PCB Design), Spice Simulation, Arduino, Raspberry Pi, Robotics Prototyping.
- **Tools & Game Dev:** Unity Game Engine, Git/GitHub, VS Code, IntelliJ IDEA, Microsoft Office, Photoshop.

Languages: Turkish (Native), English (Advanced), German (Beginner).
Interests: Swing Dance, Sailing, Rowing, Guitar Playing, Reading, Travelling.